

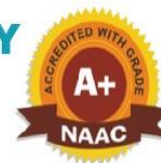


JEPPIAAR INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

"Self-Belief | Self-Discipline | Self-Respect"

Kunnam, Sunguvarchatram, Sriperumbudur – 631 604.



DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

ACBM302 - CLOUD SERVICE MANAGEMENT LABORATORY

NAME :

REG NO. :

YEAR :

SEMESTER :

BRANCH :



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BONAFIDE CERTIFICATE

This is a certified Bonafide Record Work of Mr./Ms. _____

Register No _____ submitted for the Anna University Practical Examination
held on _____ in **ACBM302 CLOUD SERVICE MANAGEMENT
LABORATORY** during the year 2026-2027.

Signature of the Lab In-charge

Head of the Department

Internal Examiner

External Examiner

INSTITUTE VISION:

Jeppiaar Institute of Technology aspires to provide technical education in futuristic technologies with the perspective of innovative, industrial and social application for the betterment of humanity.

INSTITUTE MISSION:

- ❖ **M1:** To produce competent and disciplined high-quality professionals with the practical skills necessary to excel as innovative professionals and entrepreneurs for the benefit of the society.
- ❖ **M2:** To improve the quality of education through excellence in teaching and learning, research, leadership and by promoting the principles of scientific analysis, and creative thinking.
- ❖ **M3:** To provide excellent infrastructure, serene and stimulating environment that is most conducive to learning.
- ❖ **M4:** To strive for productive partnership between the Industry and the Institute for research and development in the emerging fields and creating opportunities for employability.
- ❖ **M5:** To serve the global community by instilling ethics, values and life skills among the students needed to enrich their lives.

VISION

- ❖ To be the most preferred destination in the country for imparting education in Computer Science and Business Systems, at the undergraduate level.
- ❖ To transform learners into industry ready professionals at the global level to provide solutions for business problem and contribute to the society at large.

DEPARTMENT MISSION

- ❖ DM1: To provide an infrastructure and adopt creative teaching techniques to promote participatory learning.
- ❖ DM2: To develop high personal and professional values, business competence and a spirit of innovation and entrepreneurship.
- ❖ DM3: To encourage the desire for higher learning and research, to address global challenges.
- ❖ DM4: To collaborate with industry to inculcate varied skill sets that meets industry standards and to practice moral values.

PROGRAM EDUCATIONAL OBJECTIVES(PEO'S)

- ❖ PEO1 To ensure graduates will be proficient in utilizing the fundamental knowledge of basic sciences, mathematics, Computer Science and Business systems for the applications relevant to various streams of Engineering and Technology.
- ❖ PEO2 To enrich graduates with the core competencies necessary for applying knowledge of computer science and Data analytics tools to store, retrieve, implement and analyze data in the context of business enterprise
- ❖ PEO3 To enable graduates to gain employment in organizations and establish themselves as professionals by applying their technical skills and leadership qualities to solve real world problems and meet the diversified needs of industry, academia and research
- ❖ PEO4 To equip the graduates with entrepreneurial skills and qualities which help them to perceive the functioning of business, diagnose business problems, explore the entrepreneurial opportunities and prepare them to manage business efficiently

PROGRAM OUTCOMES

Engineering Graduates will be able to:

- ❖ PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- ❖ PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- ❖ PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- ❖ PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- ❖ PO5: Modern tool usage (k3,k5,k6) create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- ❖ PO6Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- ❖ PO7: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- ❖ PO8: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- ❖ PO9: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- ❖ PO10: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

- ❖ PO11: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PROGRAM SPECIFIC OUTCOMES

- ❖ PSO1: To create, select, and apply appropriate techniques, resources, modern engineering and business tools including prediction and data analytics to complex engineering activities and business solutions.
- ❖ PSO2: To evolve computer science domain specific methodologies for effective decision making in several critical problem domains of the real world.
- ❖ PSO3: To be able to apply entrepreneurial skills and management tools for identifying, analyzing and creating business opportunities with smart business ideas.
- ❖ PSO4: To manage complex IT projects with consideration of the human, financial, ethical and environmental factors and an understanding of risk management processes, and operational and policy implication.

TABLE OF CONTENTS

S. NO	DATE	LIST OF EXPERIMENTS	PAGE NO	MARKS	SIGN
1		Create a Cloud Organization in AWS/Google Cloud/or any equivalent Open Source cloud softwares like Openstack, Eucalyptus, OpenNebula with Role-based access control IoT Applications in Smart Pollution Control.			
2		Create a Cost-model for a web application using various services and do Cost-benefit analysis			
3		Create alerts for usage of Cloud resources			
4		Create Billing alerts for your Cloud Organization			
5		Compare Cloud cost for a simple web application across AWS, Azure and GCP and suggest the best one			

Course Outcome :

CO302.1	Create and configure a cloud organization using AWS, Google Cloud, or OpenStack/Eucalyptus/OpenNebula with appropriate users, roles, and role-based access control.	K3
CO302.2	Develop a cloud-based IoT application for smart pollution monitoring and control using appropriate cloud services.	K3
CO302.3	Develop a cost model for a web application using different cloud services and perform cost-benefit analysis.	K4
CO302.4	Configure and analyze resource-usage and billing alerts to monitor cloud consumption and control expenditure.	K3
CO302.5	Compare the cost, performance, scalability, and suitability of AWS, Microsoft Azure, and Google Cloud Platform for a simple web application and recommend an appropriate cloud provider.	K5

Mapping of Cos with Pos and PSOs

COs/ Pos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	2	1											1	1	1	1
CO2	3	2											1	1	1	1
CO3	3		1										1	1	1	1
CO4	3	2	1	1									1	1	1	1
CO5	3	2	1										1	1	1	1

1–Slight,2–Moderate,3–Substantial,BT-Bloom'sTaxonomy

Ex. No: 1
Date:

Create a Cloud Organization with Role-Based Access Control

Aim

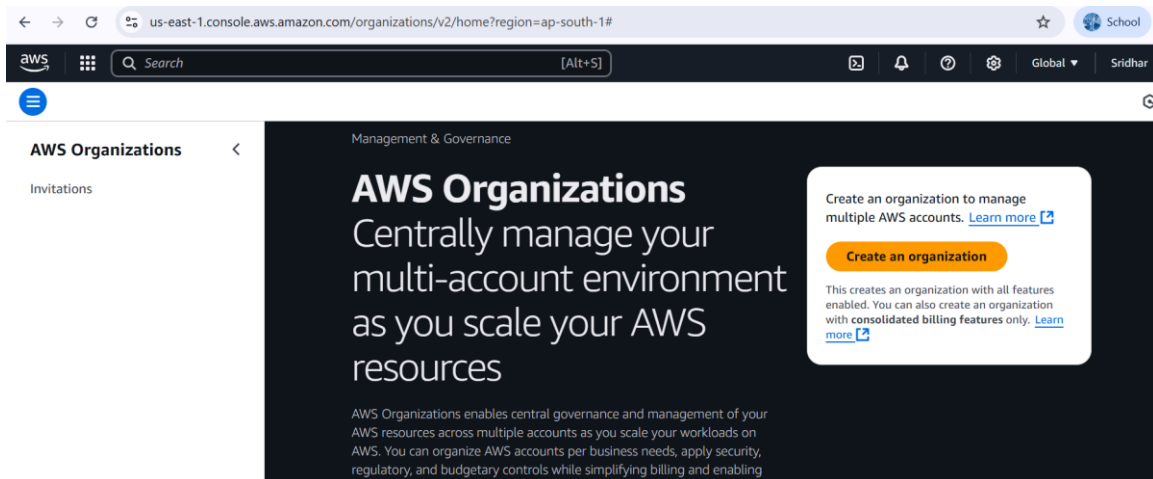
To create an AWS Organization and implement role-based access control using IAM groups, users, and policies.

Description

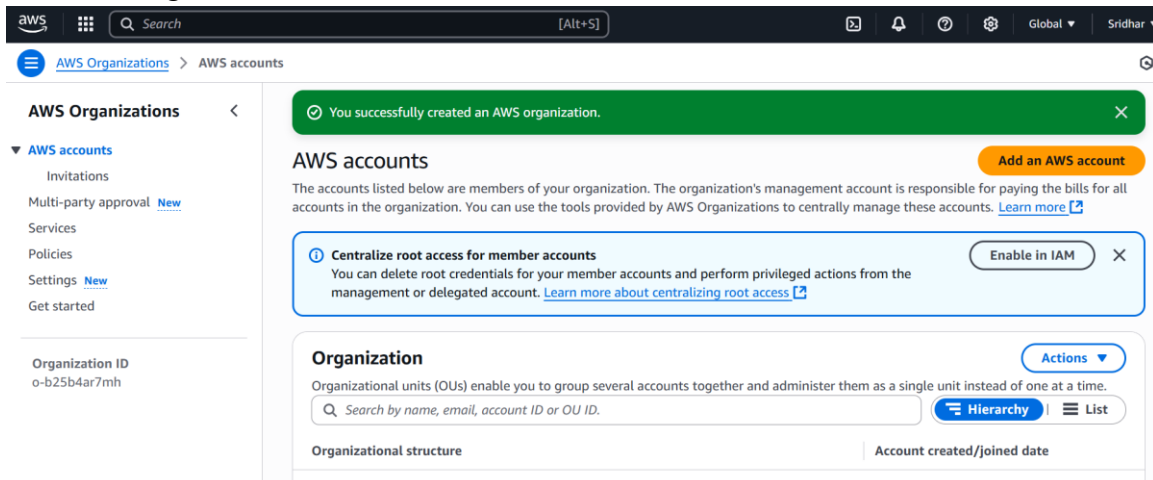
This experiment demonstrates how to manage multiple AWS accounts under one organization and implement secure access control by assigning users to roles with specific policies.

Procedure

1. Log in to AWS Console as root user at <https://aws.amazon.com>.
2. Go to AWS Organizations → Click “Create Organization” → Select “Enable All Features”.



3. Inside the org dashboard, click “Add Account” → Choose “Create an AWS Account”.



4. Enter email and account name (e.g., dev-team@example.com).

AWS Organizations > **AWS accounts** > Add an AWS account

☒ **Create an AWS account**
Create an AWS account that is added to your organization.

☐ **Invite an existing AWS account**
Send an email request to the owner of the account. The account joins the organization.

Create an AWS account

AWS account name
Sandbox

Email address of the account's owner
account@domain.com

IAM role name
The management account can use this IAM role to access resources in the member account.
OrganizationAccountAccessRole

5. Go to IAM → Click Groups → Create New Group.

User groups (0) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name Users Permissions Creation time

No resources to display

Buttons: Delete, Create group

6. Create the following groups: AdminGroup (AdministratorAccess), DevGroup (PowerUserAccess), ViewerGroup (ReadOnlyAccess).

Identity and Access Management (IAM) > **User groups** > Create user group

User group name
Enter a meaningful name to identify this group.
AdminGroup
Maximum 128 characters. Use alphanumeric and '+-._@-' characters.

Add users to the group - Optional (0) Info
An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

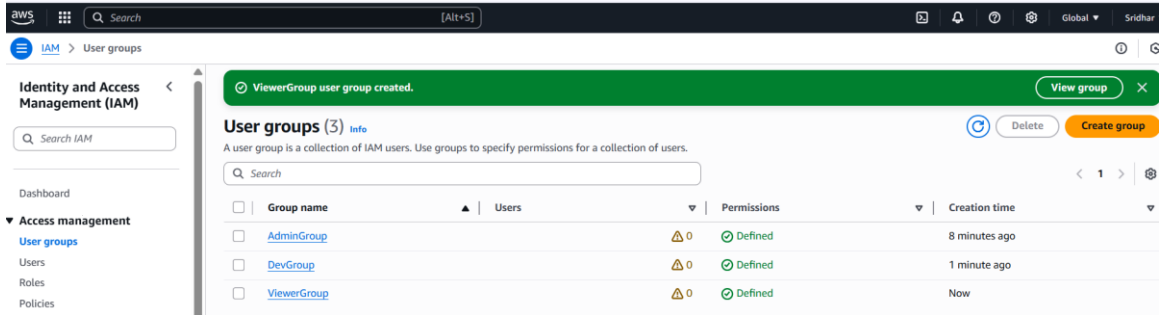
User name Groups Last activity Creation time

No resources to display

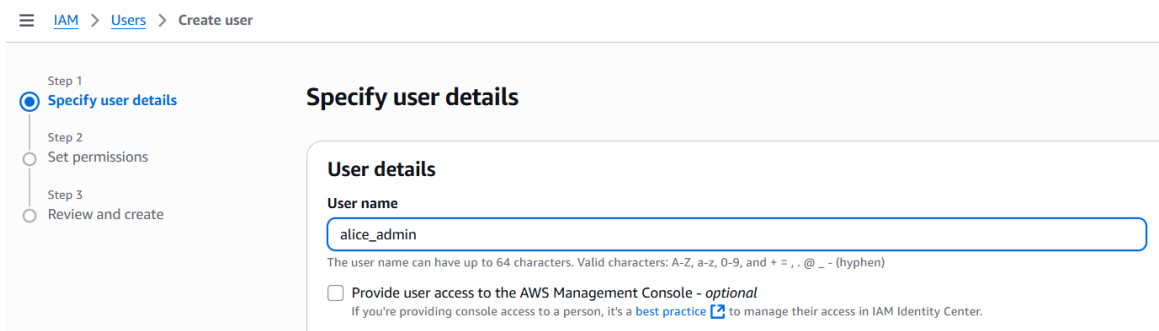
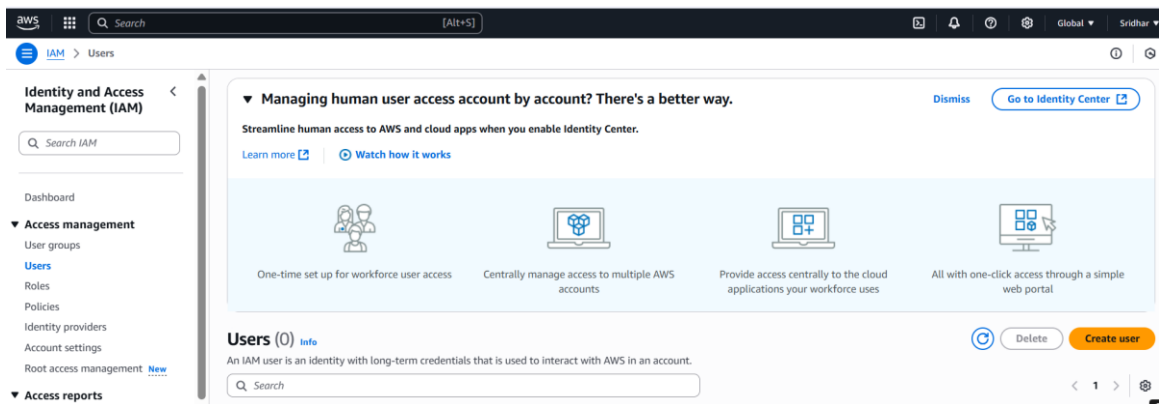
Attach permissions policies - Optional (1/1065) Info
You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Search Filter by Type: All types

Policy name	Type	Used as	Description
☒ AdministratorAccess	AWS managed - job ...	None	Provides full access to AWS services and resources.



- Go to IAM → Users → Add user: Create users: alice_admin, bob_dev, eve_viewer, enable console access.



8. Assign users to appropriate groups.

The screenshot shows the AWS IAM console 'Create user' wizard, specifically the 'Set permissions' step. The left sidebar shows the progress: Step 1 (Specify user details), Step 2 (Set permissions), and Step 3 (Review and create). The main content area is titled 'Set permissions' and includes a sub-header 'Permissions options' with two radio buttons: 'Add user to group' (selected) and 'Copy permissions'. Below this is a table titled 'User groups (1/3)' with a search bar and columns for Group name, Users, and Attached policies. The table lists three groups: AdminGroup (selected), DevGroup, and ViewerGroup. A green banner at the top of the lower section says 'User created successfully' with a 'View user' button. Below the banner are four icons representing different use cases for IAM. At the bottom, there is a 'Users (1)' section with a table showing the details of the created user, 'alice_admin'.

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

User groups (1/3)

Search

Group name	Users	Attached policies
☒ AdminGroup	0	AdministratorAccess
☐ DevGroup	0	PowerUserAccess
☐ ViewerGroup	0	AIOpsReadOnlyAccess

User created successfully
You can view and download the user's password and email instructions for signing in to the AWS Management Console. [View user](#)

[Learn more](#) [Watch how it works](#)

One-time set up for workforce user access

Centrally manage access to multiple AWS accounts

Provide access centrally to the cloud applications your workforce uses

All with one-click access through a simple web portal

Users (1) [info](#) [Delete](#) [Create user](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key
☐ alice_admin	/	1	-	-	-	-	-

9. Login as each user and verify their permissions by trying resource creation and access.

Sample Output

Organization with 2 accounts created

Users and groups with assigned roles:

Username, Group, Policy

alice_admin, AdminGroup, AdministratorAccess

bob_dev, DevGroup, PowerUserAccess

eve_viewer, ViewerGroup, ReadOnlyAccess

Result

Successfully created a multi-account AWS Organization and enforced role-based access using IAM groups and policies.

Ex. No: 2
Date:

Create a Cost-model for a Web Application and Perform Cost-benefit Analysis

Aim

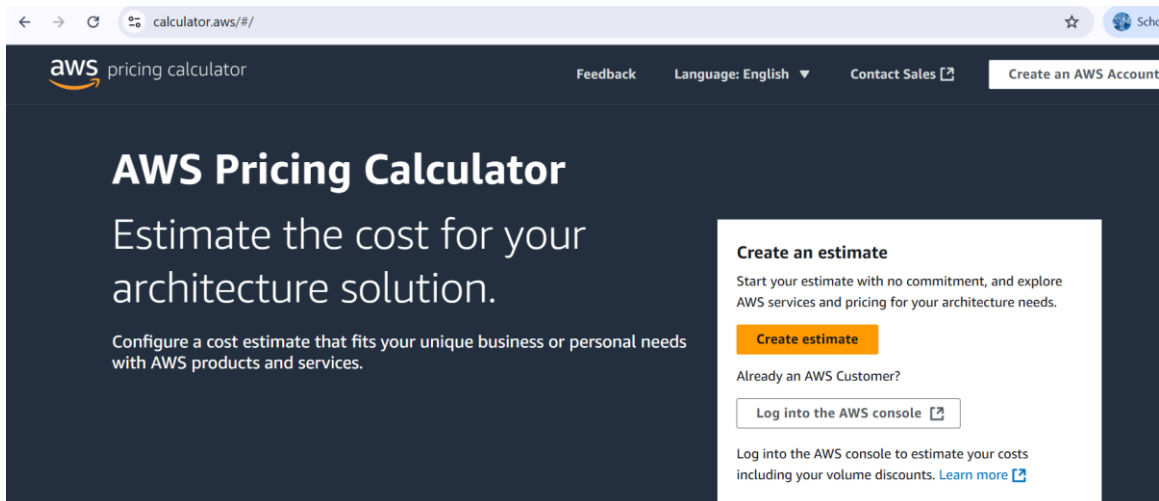
To estimate the cost of running a web application on AWS and perform a cost-benefit analysis based on chosen services.

Description

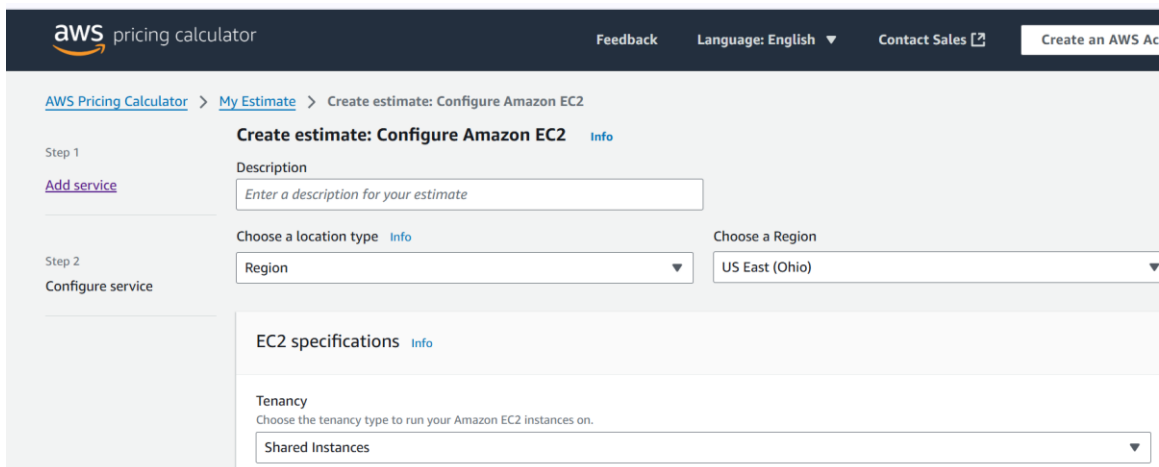
This experiment uses the AWS Pricing Calculator to model a simple web application infrastructure including EC2, S3, and RDS, and compares costs for performance optimization.

Procedure

1. Go to <https://calculator.aws/#/>



2. Click 'Create estimate'.



3. Add EC2 instance: t3.micro, Linux, 2 instances, 720 hours/month.

The screenshot shows the AWS Pricing Calculator interface for configuring EC2 instances. The table lists various instance types, with t3.micro selected. The total monthly cost is 3.80 USD.

Instance name	vCPUs	Memory	Network Performance	Storage	On-Demand Hourly Cost
☐ t3.nano	2	0.5 GiB	Up to 5 Gigabit	EBS only	0.0052
☒ t3.micro	2	1 GiB	Up to 5 Gigabit	EBS only	0.0104
☐ t3.small	2	2 GiB	Up to 5 Gigabit	EBS only	0.0208
☐ t3.medium	2	4 GiB	Up to 5 Gigabit	EBS only	0.0416
☐ t3.large	2	8 GiB	Up to 5 Gigabit	EBS only	0.0832
☐ t3.xlarge	4	16 GiB	Up to 5 Gigabit	EBS only	0.1664

Total Upfront cost: 0.00 USD
Total Monthly cost: 3.80 USD

Buttons: Show Details, Cancel, Save and view summary, Save and add service

4. Add S3 Storage: 50 GB.

5. Add RDS (MySQL): Single-AZ deployment, db.t3.micro, 20 GB storage.

The screenshot shows the AWS Pricing Calculator interface for configuring Amazon RDS for MySQL. The configuration includes a description, location type (Region), region (US East (Ohio)), and MySQL instance specifications (Nodes: 1).

Step 1: [Add service](#)

Step 2: [Configure service](#)

Create estimate: Configure Amazon RDS for MySQL

Description:

Choose a location type: [Info](#)

Region:

Choose a Region:

MySQL instance specifications [Info](#)

Nodes: Enter the number of DB instances that you need.

The screenshot shows the AWS Pricing Calculator interface for MySQL instance specifications. The top navigation bar includes the AWS logo, 'pricing calculator', and links for 'Feedback', 'Language: English', 'Contact Sales', and 'Create an AWS Account'. The main section is titled 'MySQL instance specifications' with an 'Info' link. Under the 'Nodes' section, there is a text input field with the value '1' and a search bar containing 'db.t3.micro'. A selected instance box displays 'Selected Instance: db.t3.micro', 'vCPU: 2', and 'Memory: 1 GiB'. At the bottom, the cost summary shows 'Total Upfront cost: 0.00 USD' and 'Total Monthly cost: 87.97 USD'. Action buttons include 'Show Details', 'Cancel', 'Save and view summary', and 'Save and add service'.

6. Download and review the cost summary.
7. Create a table summarizing monthly costs for each component.
8. Analyze which components contribute most to cost and evaluate alternatives.

Sample Input

Web app infrastructure with EC2, S3, RDS.

Sample Output

The screenshot shows the 'My Estimate' summary in the AWS Pricing Calculator. The top navigation bar is identical to the previous image. The 'Estimate summary' section displays 'Upfront cost: 0.00 USD', 'Monthly cost: 95.77 USD', and 'Total 12 months cost: 1,149.24 USD (Includes upfront cost)'. To the right, the 'Getting Started with AWS' section has buttons for 'Get started for free' and 'Contact Sales'. Below the summary, the 'My Estimate' section features a search bar and a table of resources. The table has columns for checkboxes, Service Name, Status, Upfront cost, Monthly cost, Description, Region, and Config Summary. Two items are listed: Amazon EC2 and Amazon RDS.

	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	7.80 USD	-	US East (Ohio)	Tenancy (Shared Instances), Operating system (Linux), Workload (Co...
☐	Amazon RDS fo...	-	0.00 USD	87.97 USD	-	US East (Ohio)	Storage amount (100 GB), Storage for each RDS instance (General P...

Result

Successfully created a cost model for a sample application and identified key cost drivers for optimization.

Ex. No: 3
Date:

Create Alerts for Usage of Cloud Resources

Aim

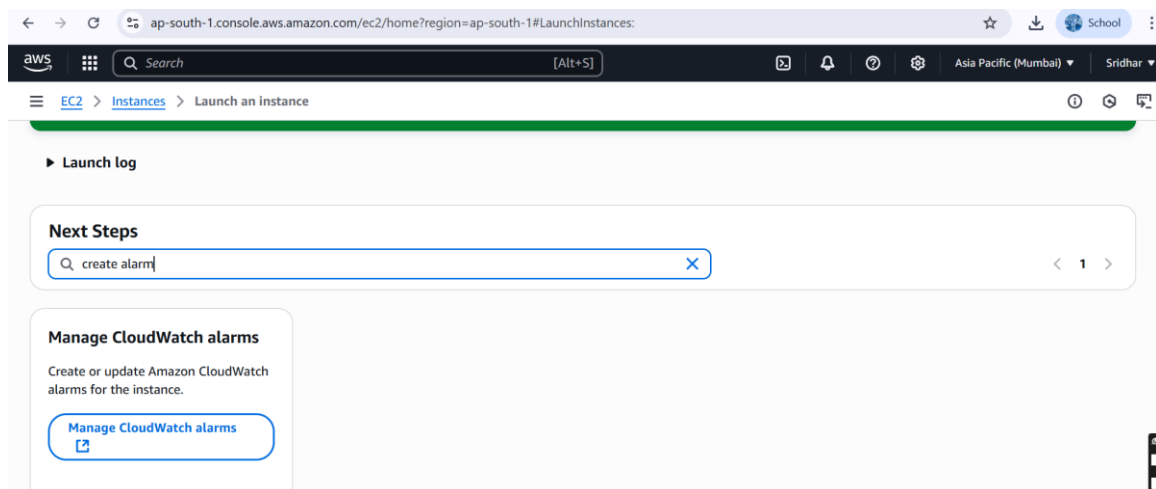
To configure AWS CloudWatch to monitor resource usage and trigger alerts when thresholds are exceeded.

Description

This experiment demonstrates setting up CloudWatch alarms for EC2 instance metrics and using SNS to notify users when thresholds are crossed.

Procedure

1. Login to AWS Console.
2. Go to EC2 → Instances → Select instance.
3. Click 'Monitor' → 'Create Alarm'.



4. Choose metric: CPUUtilization.
5. Set threshold: CPUUtilization > 70% for 5 minutes.

Alarm thresholds
Specify the metric thresholds for the alarm.

Group samples by: Average

Type of data to sample: CPU utilization

Alarm when: ≥ 70

Consecutive period: 1

Period: 5 Minutes

Alarm name: awsec2-i-Oa047e3a12b2fc3af-GreaterThanOrEqualToThreshold-CPUUtilization

Alarm description: Alarm on instance i-Oa047e3a12b2fc3af: Triggered when CPUUtilization ≥ 70 for 1 consecutive 5-minute periods.

6. Select or create an SNS topic for notification.
7. Subscribe to the topic via email and confirm subscription.
8. Test by generating high CPU load on instance.

Sample Input

Metric: CPUUtilization > 70%

Sample Output

Alarm: HighCPUAlert

Metric: CPUUtilization

Status: ALARM

Notification sent to: Sridhar.u@gmail.com

Result

Alert successfully triggered based on EC2 usage threshold; notification delivered to subscriber.

Ex. No: 4
Date:

Create Billing Alerts for Your Cloud Organization

Aim

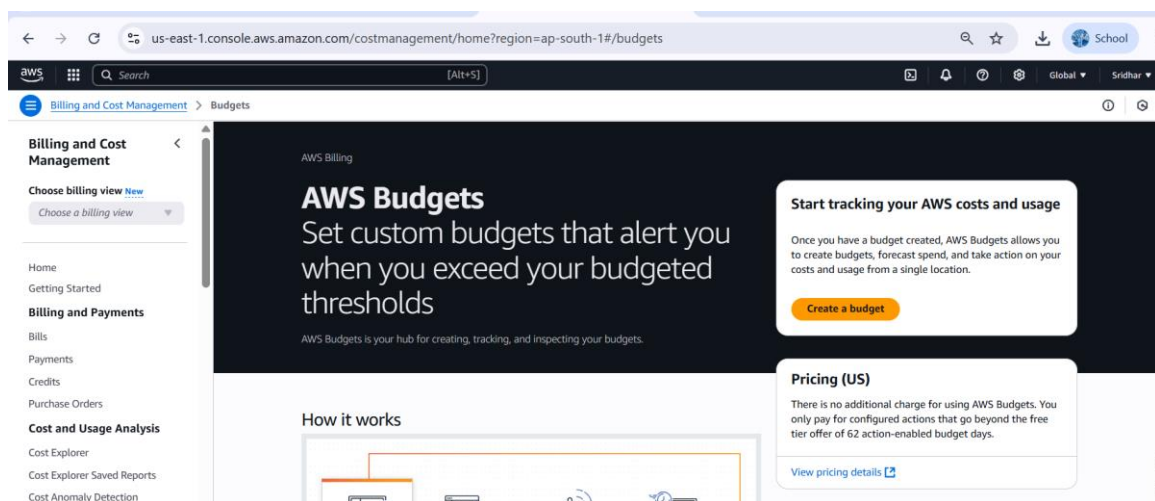
To create billing alerts that notify the user when AWS costs exceed predefined thresholds.

Description

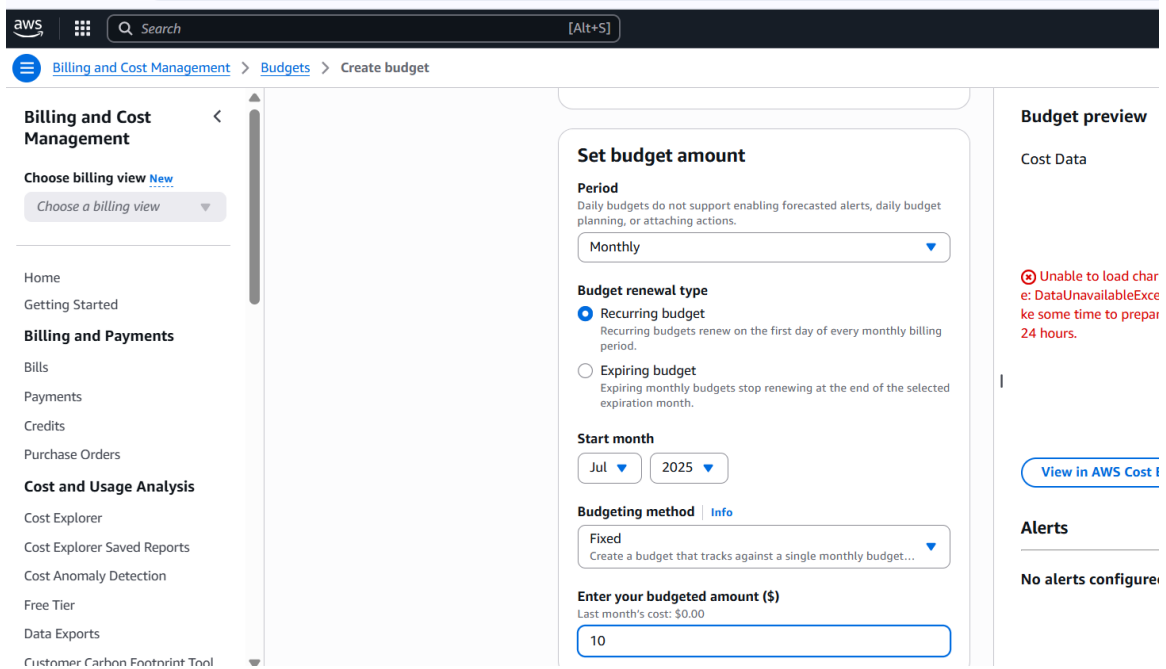
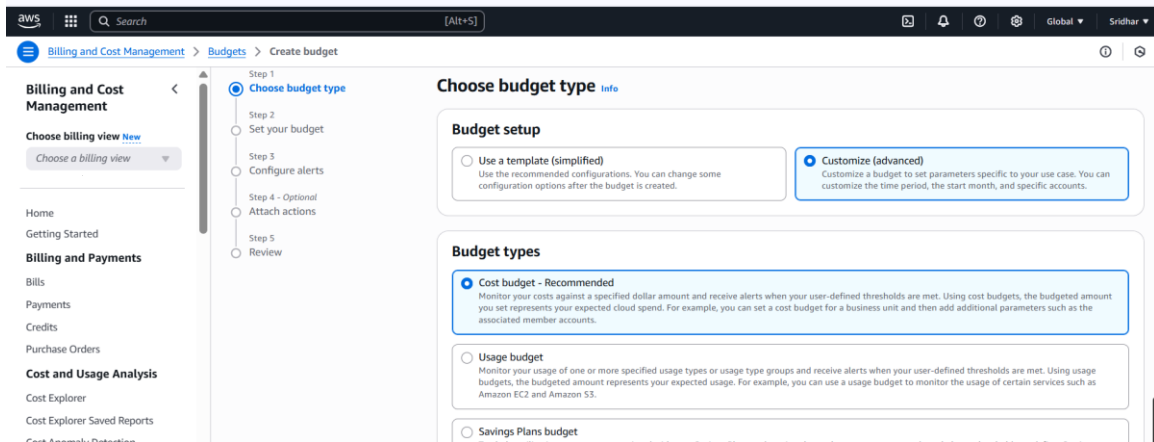
This experiment configures billing alerts using AWS Budgets to help monitor and control cloud expenses proactively.

Procedure

1. Login to AWS Console.
2. Go to Billing Dashboard → Budgets.
3. Click 'Create budget'.

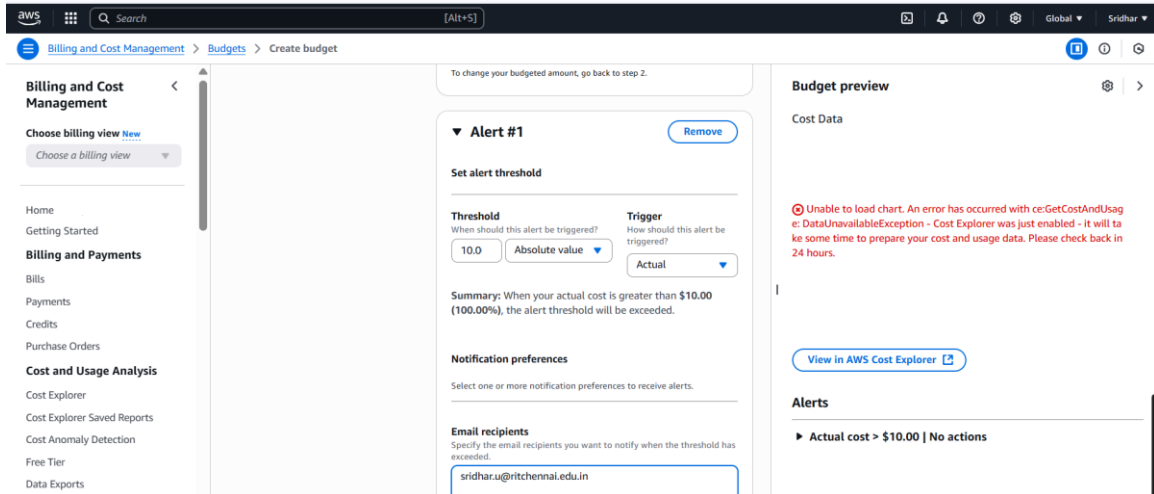


4. Choose 'Cost Budget' → Set threshold (e.g., 10 USD).

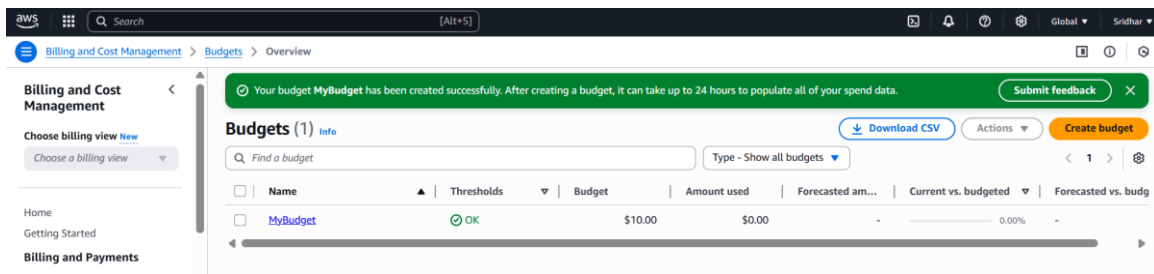


5. Set time period: Monthly.

6. Set notification when cost > threshold.



7. Choose or create an SNS topic and subscribe via email.



8. Verify that billing alert is configured and visible in budget list.

Sample Input

Threshold: 10 USD/month

Sample Output

Budget: MonthlyBudgetAlert

Threshold: \$10.00

Current Usage: \$12.43

Status: ALARM

Notification sent to: billing-sridhar.u@ritchennai.edu.in

Result

Billing alert configured successfully; alert triggered when usage exceeded \$10.

Ex. No: 5

Date:

Compare Cloud Cost for a Simple Web App Across AWS, Azure, and GCP

Aim

To compare cloud hosting costs across three major providers and suggest the most cost-effective solution.

Description

This experiment estimates costs of hosting a basic web app (2 VMs, 50 GB storage, 1 database) across AWS, Azure, and GCP and provides a comparative analysis.

Procedure

1. Use AWS Pricing Calculator to estimate cost for: EC2, S3, RDS.
2. Use Azure Pricing Calculator for: Virtual Machines, Blob Storage, SQL Database.
3. Use GCP Pricing Calculator for: Compute Engine, Cloud Storage, Cloud SQL.
4. Export and compile the results in a spreadsheet.
5. Compare features, SLA, and support options for each provider.
6. Recommend the most suitable provider based on cost and performance.

Sample Input

Specs: 2 VMs, 50 GB storage, 1 DB (monthly usage)

Sample Output

Provider, MonthlyCostUSD, SLA, Notes

AWS, 42.20, 99.99%, Scalable and reliable

Azure, 44.10, 99.9%, Windows integration

GCP, 38.50, 99.95%, Most cost-effective

Result

GCP recommended as the most economical provider for the sample web application.